Aditya

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PROFESSIONAL PROFILE

Analytical Computer Science & Data Analytics student at IIT Patna with expertise in Machine Learning, predictive modeling, and scalable software architecture. Proven track record in building end-to-end data systems, ranging from automated retail demand forecasting to high-dimensional time-series classification. Adept at translating complex datasets into actionable business intelligence using Python, modern ML stacks, and enterprise-grade data engineering pipelines.

EDUCATION

Indian Institute of Technology, Patna May 2029 (Expected)
Bachelor of Science in Computer Science & Data Analytics
Relevant Coursework: Probability & Statistics, Time-Series Analysis, Linear Algebra, Machine Learning, Data Structures & Algorithms, Financial Modeling.

TECHNICAL SKILLS

- **Data Science & AI:** Python (Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow, Keras, Prophet), Time-Series Forecasting, Regime Detection (HMM, KMeans), Statistical Modeling.
- **Software & Data Engineering:** Modular System Design, SQL, Git, Apache Airflow (ETL Pipelines), Postgres, Vectorized Computation, API Development.
- **Visualization & Analytics:** Streamlit, Plotly, Matplotlib, React, Performance Attribution, Risk Analytics (Sharpe Ratio, Max Drawdown).
- **Domain Expertise:** Supply Chain Optimization, Financial Analytics, Market Microstructure, Systematic Decision Support.

KEY PROJECTS

Adaptive AI-Driven Time-Series Classification System *Python, XGBoost, HMM, Streamlit*

- Engineered a production-grade context-detection framework using Hidden Markov Models (HMM) and KMeans to classify high-dimensional time-series data into distinct operational regimes.
- Developed an XGBoost-based predictive model to forecast state transitions with 65-75% accuracy, enabling dynamic resource allocation and proactive risk-mitigation strategies.
- Optimized system performance to process 17+ years of large-scale datasets, achieving a 56% improvement in risk-adjusted performance metrics compared to static baseline models.
- Architected a modular, ”time-series-aware” ML pipeline with lagged features to eliminate look-ahead bias and ensure 100% institutional-grade research reproducibility.
- Built an interactive Streamlit dashboard for real-time visualization of model performance, rolling risk profiles, and system heatmaps.

Enterprise Retail Demand Forecasting & Inventory Optimization *Python, Prophet, Airflow, Postgres*

- Developed an end-to-end ML system to predict next-week sales at the SKU level, significantly reducing overhead costs associated with overstock and stockouts.
- Built automated data ingestion and transformation pipelines using Apache Airflow to process large-scale retail datasets while accounting for seasonality and trend detection.
- Deployed a hybrid forecasting model (Prophet + XGBoost) to generate high-precision demand forecasts integrated with a rule-based reorder recommendation engine.
- Designed a full-stack dashboard (Streamlit/React) for store managers to visualize stock-level alerts, forecasted trends, and automated replenishment insights.
- Validated system robustness through backtesting on historical inventory data, evaluating model reliability across diverse regional market environments.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **CFA Program: Level I Candidate** (Chartered Financial Analyst Institute)
- **NISM Series VIII: Equity Derivatives Certification** (National Institute of Securities Markets)

RESEARCH INTERESTS

Applied Machine Learning, Market Microstructure, Systematic Forecasting, Risk Management, Supply Chain Intelligence, Scalable Data Systems.